Nonlinear Oscillations in Chemical and Biological Systems

On the search of a chemical oscillator

On mathematics and the life sciences

You have probably either heard or seen a lot of mathematics modelling in the usual fields: physics, economics, and everything that requires statistics. Mathematics has been foundational to the way we understand physics, and has been so for centuries. I want to take the next three weeks to (hopefully) make you think about mathematics instead as a tool for the life sciences: everything from biology to chemistry and medicine. We will not have time to cover every single one of these fields, but rest assured that all the techniques that are covered in these notes can also be applied to those fields.

The most common tool of the mathematician working on a foreign field is the model. What is a model? Maybe it is 'a description of an experimentally delineated phenomenon by means of mathematics' (van den Berg, 2011), maybe it is 'a caricature of a real system (Segel and Edelstein-Keshet, 2013)', or maybe it is 'a tool to describe a natural phenomenon' (Small, 2011). What all these descriptions have in common is the idea that models are merely abstract ideals of things that exist out there in the real world.

Why model? It's a question of complexity vs accuracy. Natural phenomena are incredibly complicated and hard to understand. Indeed, if we could completely comprehend the way the ocean current describes the movement of the clouds, we could not need to model the weather. But alas, we cannot. The next best thing we can do is then to create a model of the phenomenon we are trying to study. But how to make this choice? One might think that increasing the complexity of the model (by adding more variables or increasing the complexity of the functions involved) is the obvious solution.

On one hand, adding complexity for its own sake does not immediately make a model more fateful. Additionally, the more we specify the model the less generic it becomes. One of the advantages of working with models (as we will see in the coming weeks) is that by nature of being abstract mathematical objects, they are not attached to the initial phenomenon we wanted to study. A model of electrical conduction in a circuit drafted by an engineer may also be used to model the behaviour of neurons; a model of interaction between molecules created by a chemist might also be used in biology to explain predator-prey dynamics.

On the other extreme of the spectrum, making a model too simple risks making it a 'toy' model, as the experts describe it, too far away from reality to be useful in the real world.

This is not to say that toy models are useless, on the contrary, they are perfect as starting points to create more complex models; or sometimes they can be handed over to the pure mathematicians to be used to create new theory.

In short, to quote the old adage, 'all models are wrong, but some are useful'.

Exercise 1:

Read chapter 5 of van den Berg (2011), the author delves into a much more thorough discussion of 'Good modelling practice' than we can get into at this time.

When used correctly, models can sometimes even inform the original field of study. And, in a few selected cases, they can even predict a discovery that the experts on that field were not expecting (or outright refusing to believe) at the time.

Oscillators in nature

Does a plant know when it is daytime? A sailor by the name of Androstenes of Tasos was the first person in recorded history to pose that question. He was part of Alexander the Great's fleet, and in one of his travels across the African coast he noted that the leaves of the tamarind trees opened up during daytime and closed off at dusk. At the time, Androstenes concluded that this was a passive response to the appearance of daylight, and this view would become common for centuries after he passed away (Edelstein-Keshet, 2005). Today, this phenomenon now receives the name of 'nyctinasty', and is one of the many examples of a biological oscillator.

Biological oscillators are among the most studied biological systems and include

- Circadian rhythms (i.e., 'nyctinasty').
- The cell cycle (i.e., cellular reproduction).
- Periodic hormonal rhythms.
- Intracellular oscillations in the concentrations of a variety of chemical species; calcium, for instance.
- Rhythms in predator-prey systems.
- Oscillations in stirred chemical reactors.
- Neuronal rhythms in the brain.

One key feature of many of these oscillating phenomena is their *periodicity*, the quality of returning to the initial state after a fixed period of time. We can see the sun rising every morning, hear the cicadas crying out every few years, and feel the horrible cold of the winter weather every few months.

The rest of this course will focus on a particular type of oscillators, but I want to restate here that none of the techniques which we will be studying in the coming weeks are exclusive to this field.

But before we move forward into the world of oscillations, we first need to introduce a very important tool that will facilitate the modelling process

The Law of Mass Action

This fundamental 'law' of chemistry describes the rate at which chemicals, whether large macromolecules or simple ions, collide and interact to form different chemical combinations. Suppose that two compounds, say A and B, react upon collision with each other to form product C,

$$\mathrm{A} + \mathrm{B} \underset{k_-}{\overset{k_+}{\rightleftarrows}} \mathrm{C}, \tag{1}$$

The Law of Mass Action states that '*[. . .] the rate of reaction is proportional to the concentrations of the two reactants*' (Segel and Edelstein-Keshet, 2013). Thus, the *forward* reaction rate (also known as *association* rate) is given by $k_+[A][B]$ (where [A] denotes the concentration of A, and [B] the concentration of B). Analogously, the backwards (*disassociation*) rate is given by $k_-[C]$

A special (and very common) type of reaction is *dimerisation*, when two of the same molecules react with each other, which is written as

$$\mathrm{A} + \mathrm{A} \underset{k_-}{\overset{k_+}{\rightleftarrows}} \mathrm{C}. \tag{2}$$

In this case, the forward reaction rate is given by $k_+[\mathrm{A}][\mathrm{A}] = k_+[\mathrm{A}]^2$.

Exercise 2:

A fundamental reaction scheme used to study the kinetics of enzymes is given by

$$\mathrm{E} + \mathrm{S} \underset{k_{-1}}{\overset{k_1}{\rightleftarrows}} \mathrm{C} \xrightarrow{\mathrm{k_2}} \mathrm{E} + \mathrm{P}. \tag{3}$$

Where an Enzyme (E) causes a Substrate (S) to turn into a Product (P). This transformation is a two-step process, which must first undergo a transformation into an intermediate Complex (C). As a warm-up, calculate all the forward and backward reaction rates involved in this example.

With this brief introduction of the Law of Mass Action in mind, let us go back to the story of one of the first models of an oscillator.

The impossibility of a homogeneous oscillator

Alfred J. Lotka was a biochemist, biomathematician and biostatistician in the early XX century (back when you could be all of those at the same time). Around 1910 he proposed a model the following model of chemical reactions based on the law of mass action (Lotka, 1909)

$$A \to X, \tag{4a}$$
$$X + Y \to 2Y, \tag{4b}$$
$$Y \to P. \tag{4c}$$

This model can be broken down into its components:

- The *intermediate species* (AKA the important ones): X and Y. These are the variables that we want to pay attention to. (These days we would think of them as 'predator' and 'prey', but that comes later).

- The *constant species* (AKA the non-important ones): A and B. Their existence is necessary so to control the concentrations of X and Y, but usually we consider them to be in such great quantity (in comparison to X and Y) that they might as well be constant. [1]

Truth be told, this model did not become an international superstar. It did not even really become famous until Lotka changed equation (4a) to $A + X \to 2X$. This reaction is *autocatalytic* (i.e., X is both reactant and product). This change resulted in the more 'classical' for of the Lotka system, given by

$$A + X \to 2X, \tag{5a}$$
$$X + Y \to 2Y, \tag{5b}$$
$$Y \to P. \tag{5c}$$

We will study this system in depth in just a moment, but there is some necessary context here before we move forward. Lotka initially developed his model to study autocatalytic reactions, and his model (as we are about this find out) is one of the first ones to present a **periodic solution** (an oscillator, in other terms). However, the chemists at the time were not too thrilled by the prospect. In short, by the 1920s there was no experimental evidence of an oscillating chemical reaction. The community at the time thought that Lotka's models were a gross simplification of existing phenomena, and even worse, there was a belief that they violated the Second Law of Thermodynamics (which is a *big no no*, believe me) (Zhabotinsky, 1991; Epstein and Showalter, 1996). In contrast, the model was much more well received by the biologists of the time, which is why this Lotka's (Lotka-Volterra, to be exact) model is more know as the 'predator-prey' model these days (It's just as lousy as a predator-prey model as it is for a chemical reaction model, but again, I am getting ahead of myself).

Mathematical analysis

Time to get into the mathematics part of this course. First, let us go back to the toy example of the Law of Mass Action

$$\mathrm{A} + \mathrm{B} \underset{\mathrm{k_-}}{\overset{\mathrm{k_+}}{\rightleftharpoons}} \mathrm{C}. \tag{6}$$

(This reaction is actually two reactions in a trenchcoat) Given that [A] describes the concentration of A, it is natural to ask: what is the value of [A] at any given moment during the reaction? The forward reaction occurs at the rate k_+, multiplied by the concentration of the necessary reactants, in this case A and B, in other words

$$k_+[\mathrm{A}][\mathrm{B}]. \tag{7}$$

The backwards reaction occurs at a rate k_-, and is proportional to the concentration of the reactant, C in this case. This is written as

$$k_-[\mathrm{C}] \tag{8}$$

[1] Lotka himself supposed that A and B were introduced or removed from the system as necessary to keep the reaction going.

The rate of change of [A] is then adding these two rates together

$$\frac{d[\mathrm{A}]}{dt} = k_-[\mathrm{C}] - \mathrm{k}_+[\mathrm{A}][\mathrm{B}]. \tag{9}$$

The rate of change of [C] is similar, but different, since the forward reaction instead increases the concentration of C and the backward reaction lowers it. Thus

$$\frac{d[\mathrm{C}]}{dt} = k_+[\mathrm{A}][\mathrm{B}] - \mathrm{k}_-[\mathrm{C}]. \tag{10}$$

At equilibrium concentrations are not changing, so

$$\frac{d[\mathrm{A}]}{dt} = 0, \tag{11a}$$

$$k_-[\mathrm{C_{eq}}] - \mathrm{k}_+[\mathrm{A_{eq}}][\mathrm{B_{eq}}] = 0, \tag{11b}$$

$$k_-[\mathrm{C_{eq}}] = k_+[\mathrm{A_{eq}}][\mathrm{B_{eq}}]. \tag{11c}$$

Which gives us the identity

$$\frac{k_-}{k_+} \equiv K_{\mathrm{eq}} = \frac{[\mathrm{A}]_{\mathrm{eq}}[\mathrm{B}]_{\mathrm{eq}}}{[\mathrm{C}]_{\mathrm{eq}}}. \tag{12}$$

The ratio k_-/k_+, denoted by K_{eq}, is called the *e*quilibrium constant of the reaction. It describes the relative preference for the chemicals to be in the combined state C compared to the disassociated state. If K_{eq} is small, then at steady state most of A and B are combined to give C.

Assuming that there are no other reactions involving A and C, then $[\mathrm{A}] + [\mathrm{C}] = \mathrm{A}_0$ is constant, and

$$[\mathrm{C}]_{\mathrm{eq}} = \mathrm{A}_0 \frac{[\mathrm{B}]_{\mathrm{eq}}}{\mathrm{K_{eq}} + [\mathrm{B}]_{\mathrm{eq}}}, \qquad [\mathrm{A}]_{\mathrm{eq}} = \mathrm{A}_0 \frac{\mathrm{K_{eq}}}{\mathrm{K_{eq}} + [\mathrm{B}]_{\mathrm{eq}}}. \tag{13}$$

Thus, when $[\mathrm{B}]_{\mathrm{eq}} = \mathrm{K_{eq}}$, half of A is in the bound state at equilibrium.

Exercise 3:

Consider the dimerisation reaction given by

$$\mathrm{A} + \mathrm{A} \underset{k_-}{\overset{k_+}{\rightleftarrows}} \mathrm{C}. \tag{14}$$

Derivate the differential equation for [A], the equilibrium constant of the reaction, and the conserved quantity in this case.

Time to apply this principle to the Lotka model. If you have been paying attention, you will notice that the equations are lacking reaction rates. It is also a perfect time to present the equations again, this time in its more complete form:

$$\mathrm{A} + \mathrm{X} \xrightarrow{\mathrm{a_1}} 2\mathrm{X}, \tag{15a}$$

$$\mathrm{X} + \mathrm{Y} \xrightarrow{\mathrm{a_2}} 2\mathrm{Y}, \tag{15b}$$

$$\mathrm{Y} \xrightarrow{\mathrm{b_1}} \mathrm{B}. \tag{15c}$$

Using the law of mass action on (15) results in a system of ordinary differential equations (ODEs) given by

$$\frac{d[\mathrm{X}]}{dt} = a_1[\mathrm{A}][\mathrm{X}] - a_2[\mathrm{X}][\mathrm{Y}], \tag{16a}$$

$$\frac{d[\mathrm{Y}]}{dt} = a_2[\mathrm{X}][\mathrm{Y}] - b_1[\mathrm{B}][\mathrm{Y}], \tag{16b}$$

with five paramters a_1, a_2, b_1, [A] and [B]. An easy way to reduce the number of parameters in this system is by rescaling

$$a_1 := a_1[\mathrm{A}] \qquad\qquad b_1 := b_1[\mathrm{B}] \tag{17}$$

(in reality we would have to define new parameters, but who has time for that). To simplify notation, we can also drop the concentration notation for now and present the model as

$$\frac{d\mathrm{X}}{dt} = a_1\mathrm{X} - a_2\mathrm{XY}, \tag{18a}$$

$$\frac{d\mathrm{X}}{dt} = a_2\mathrm{XY} - b_1\mathrm{Y}, \tag{18b}$$

which is also the more 'common' version of the equations.

Phase-plane analysis

ODEs are a fantastic tool from a modelling point of view, as you probably noticed from how straightforward it was to translate a chemical reaction into a mathematical expression. Unfortunately, analysing the solutions of a system of ODEs is... less straightforward. That is partially because, in most cases, writing down the expressions for the solutions as time series is down-right impossible; and in the cases where it is possible, they tend to be so complicated that it is hard to elucidate a lot of information from them.

For this reason, a lot of the dynamical systems theory around ODEs is focused more on *qualitative* elements, instead of the hard *quantitative* properties. Think of this duality as the difference between describing the shape of a curve in terms of how it 'looks' and writing down the equation that describes it.

Phase plane analysis is a very useful qualitative tool to describe the behaviour of the solution of a system like (18) without having to find the solution to the corresponding equations. In case you haven't encountered these concepts before, solutions of a system of ODEs are usually presented in terms of time-series (i.e., X vs t). For the purposes of the analysis it makes more sense to think of these solutions in the $X-Y$ plane, otherwise known as the *phase plane*.

A phase plane is a canvas, and we must populate it with information before it becomes useful. First, we can sketch the curves which satisfy either $\frac{dX}{dt} = 0$ or $\frac{dY}{dt} = 0$, these are given by

$$\frac{d\mathrm{X}}{dt} = 0, \tag{19a}$$

$$a_1\mathrm{X} = a_2\mathrm{XY}, \tag{19b}$$

and

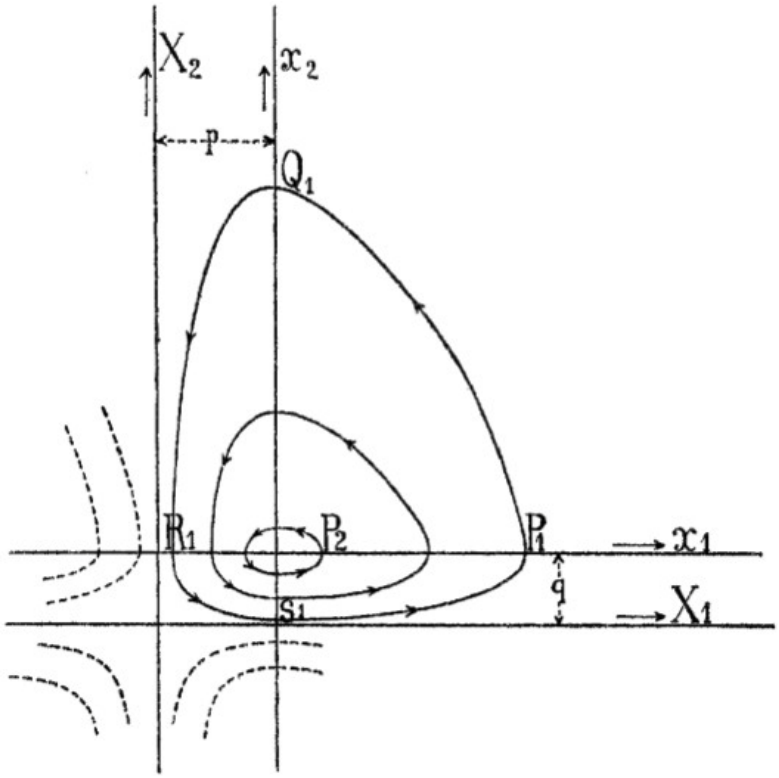


Figure 1: Phase plane of the Lotka model, stolen from the original paper by Lotka (Lotka, 1909).

$$\frac{d\mathrm{Y}}{dt} = 0, \tag{20a}$$
$$a_2\mathrm{XY} = b_1\mathrm{Y}. \tag{20b}$$

These correspond to the straight lines you can see in Figure 1 (although Lotka called his variables X_1 and X_2)

As a reminder, the **steady states** are the points (typically) at which the whole system equals 0 (i.e., the variables stop moving). They are also the intersection of the nullclines!

A small reminder on linearisation

Generically, it is very difficult to study the trajectories of a non-linear ODE system (and very trivial to study a linear system). Steady states are very important to qualitative analysis since, roughly speaking, they tend to define the landscape of the phase plane in a neighbourhood around them (how 'nearby' is this nearby? Don't ask.). In situations where we cannot find the 'true' behaviour of the system around a steady state, we can always resort to the next best think: approximate.

Fortunately for us, since at a steady state $(\mathrm{X}^*, \mathrm{Y}^*)$

$$\frac{dX}{dt} = f(X, Y) = 0, \tag{21a}$$
$$\frac{dY}{dt} = g(X, Y) = 0, \tag{21b}$$

we can make use of the **linearisation** trick. Recall that, if f and g are 'nice' functions we can approximate them by

$$f(\mathrm{X}^* + x, \mathrm{Y}^* + y) = f(\mathrm{X}^*, \mathrm{Y}^*) + \left.\frac{\partial f}{\partial X}\right|_{\mathrm{X}^*,\mathrm{Y}^*}(x) + \left.\frac{\partial f}{\partial Y}\right|_{\mathrm{X}^*,\mathrm{Y}^*}(y), \tag{22a}$$
$$g(\mathrm{X}^* + x, \mathrm{Y}^* + y) = g(\mathrm{X}^*, \mathrm{Y}^*) + \left.\frac{\partial g}{\partial X}\right|_{\mathrm{X}^*,\mathrm{Y}^*}(x) + \left.\frac{\partial g}{\partial Y}\right|_{\mathrm{X}^*,\mathrm{Y}^*}(y). \tag{22b}$$

Since the functions vanish at the steady state, we can approximate the whole system (nearby the steady state) by a much simpler linear system given by

$$\frac{dX}{dt} \approx aX + bY, \tag{23a}$$
$$\frac{dY}{dt} \approx cX + dY, \tag{23b}$$

where

$$\begin{aligned} a &= \left.\tfrac{\partial f}{\partial X}\right|_{\mathrm{X}^*,\mathrm{Y}^*}, \quad b = \left.\tfrac{\partial f}{\partial Y}\right|_{\mathrm{X}^*,\mathrm{Y}^*}, \\ c &= \left.\tfrac{\partial g}{\partial X}\right|_{\mathrm{X}^*,\mathrm{Y}^*}, \quad d = \left.\tfrac{\partial g}{\partial X}\right|_{\mathrm{X}^*,\mathrm{Y}^*}. \end{aligned} \tag{24}$$

Which you might recognise as the **Jacobian** of the system. From here, we can look at the stability of the linear system and use that to approximate the stability of the full system. I will skip over a lot of the details of this part in these notes, but you can have a read of Segel and Edelstein-Keshet (2013) for a more in-depth explanation.

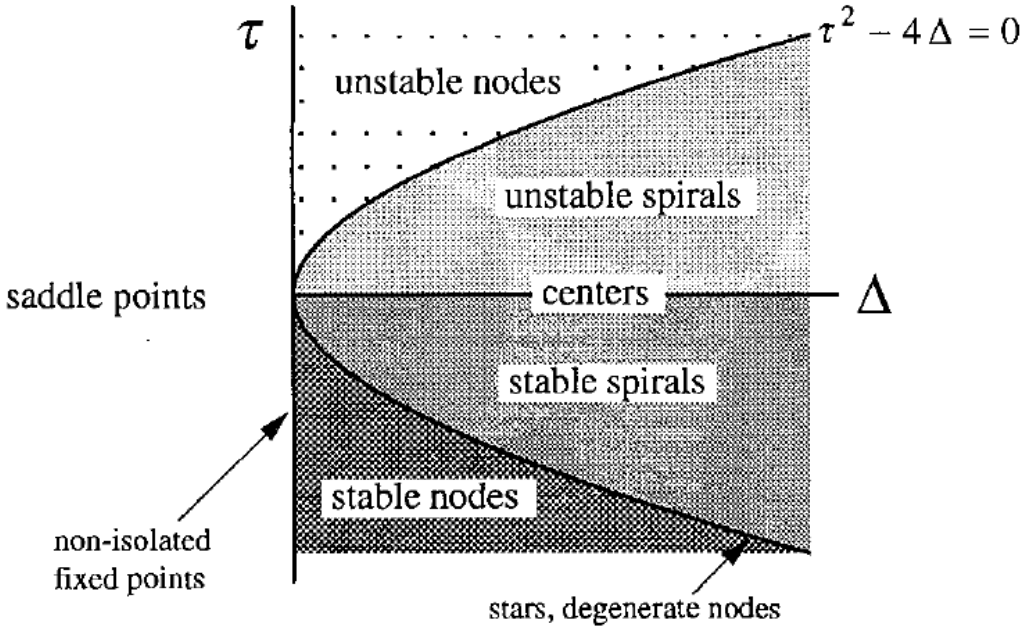


Figure 2: The trace (τ)-determinant (Δ) plane, used to classify the possible behaviours of the trajectories near the steady state, and shamelessly stolen from Strogatz (2019).

The Lotka model was two possible steady states:

$$(\mathrm{X}_0, \mathrm{Y}_0) = (0,0) \qquad \text{or} \qquad (\mathrm{X}^*, \mathrm{Y}^*) = \left(\frac{b_1}{a_2}, \frac{a_1}{a_2}\right). \tag{25}$$

With the stability matrix having the form

$$J(\mathrm{X}, \mathrm{Y}) = \begin{pmatrix} a_1 - a_2\mathrm{Y} & -a_2\mathrm{X} \\ a_2\mathrm{Y} & a_2\mathrm{X} - b_1 \end{pmatrix}. \tag{26}$$

Let's do the origin first. The local stability matrix at $(0,0)$ has the form

$$J(0,0) = \begin{pmatrix} a_1 & 0 \\ 0 & -b_1 \end{pmatrix}. \tag{27}$$

As a refresher, the stability of the steady state can be inferred by the signs of the **eigenvalues** of the Jacobian. You can calculate these values by solving the characteristic polynomial of the system, or by using the trace-determinant plane (if you have come across that concept).

For the origin, the eigenvalues are $\lambda_1 = a_1$ and $\lambda_2 = -b_1$. The origin is a **saddle point**, with trajectories approaching the steady state in one direction, and leaving it in a different direction.

Exercise 4:

The instability of the origin means that a trajectory in this system could, in principle, cross either of the axes. This could be *terrible news* for a system trying to model

concentrations of chemicals / populations / etc. Why? To make sure that this problem does not bother us in this system, find the corresponding *eigenvectors* for the origin and make sure they do not point in the wrong direction.

Local stability matrix at $(\mathrm{X}^*, \mathrm{Y}^*)$ is

$$J(\mathrm{X}^*, \mathrm{Y}^*) = \begin{pmatrix} 0 & -b_1 \\ a_1 & 0 \end{pmatrix}. \tag{28}$$

This time the eigenvalues are complex number with zero real part. This is also known as bad news, since it is the one case where our linearisation trick does not give that much information.

Exercise 5:

As a refresher, compute the characteristic polynomial of both steady states and give the explicit form of the eigenvalues. You won't have to do it every time, but it is good to do it at least once in your life.

Take a deep breath now. What was the whole point of this analysis? Lotka introduced this model to study autocatalytic reactions and oscillators. So, where are the oscillations? Normally, here is where we would use the steady states to try to find more information about the existence of periodic solutions, but the linearisation wasn't particularly helpful. The other thing you can do in a situation like this is to numerically compute some solutions to the system, like in Figure 3.

Observe the equations for the system once again. There is one 'quirky' trick we can use in this particular system. Divide one equation by the other to get

$$\frac{dX}{dY} = \frac{a_1 X - a_2 XY}{a_2 XY - b_1 Y}, \tag{29}$$

from which it follows that

$$\frac{(a_2 X - b_1)dX}{X} = \frac{(a_1 - a_2 Y)dY}{Y}. \tag{30}$$

Thus

$$a_2 X - b_1 \ln X = a_1 \ln Y - a_2 Y + K, \tag{31}$$

for some constant K that is determined by the initial conditions.

Every solution for positive X and Y is a closed curve in the (X, Y) plane. See Fig. 1. The solutions go around this curve in a counterclockwise direction. You can work this out by looking at the sign of $\dot{X}$ and $\dot{Y}$ as X and Y are bigger or smaller than the non-trivial steady state. For example, $X < p, Y < q \Rightarrow \dot{X} > 0, \dot{Y} < 0$. Etc. Actually, that one example is enough to tell us the direction of movement.

This is a lousy model for oscillatory chemical reactions, as the quadrant is **foliated** with periodic orbits. Any slight perturbation puts you on a different orbit, and which orbit you end up on depends on the initial conditions.

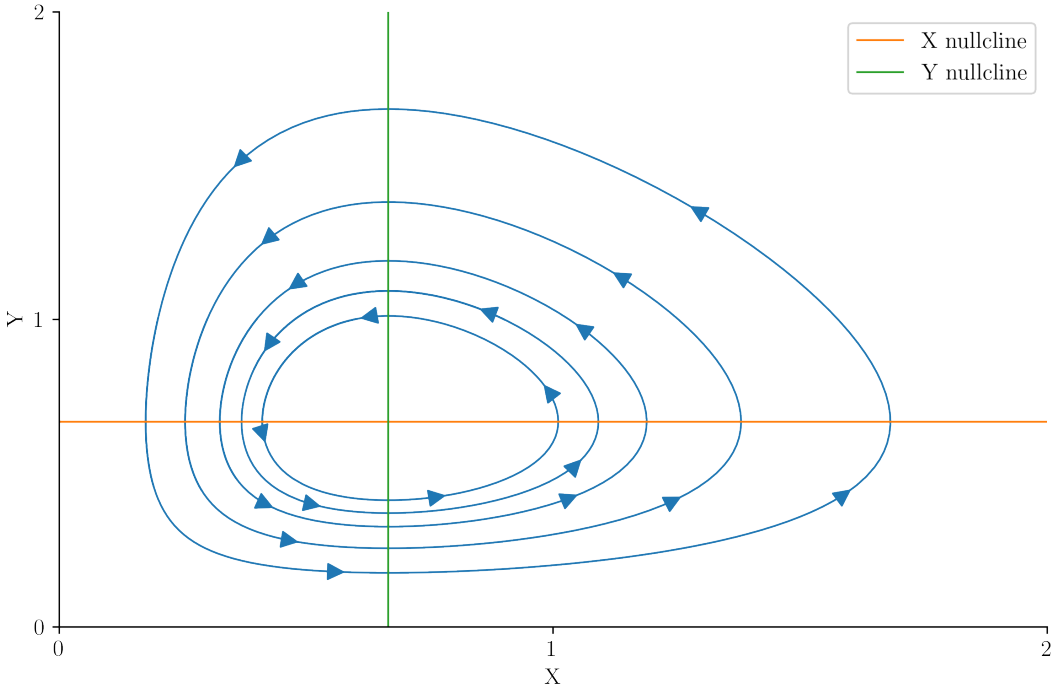


Figure 3: Some periodic orbits exhibited by the lotka model, alongside the nullclines.

Exercise 6:

We found our periodic solutions! That's great, right? Have a moment to think about the particular solutions we found and the qualities I just described. Do we want these properties? Try doing an online search for the term **structural stability**.

The Hirniak model

Let us go back to the XX century. Around the same time that Lotka presented his original model many other scientists were trying to either find them experimentally or create model that could give 'hints' to find the elusive chemical oscillation. Truly, interest in such phenomena has peaked in the first years of the new century. At the time, there was no 'well-formed' theory that described such oscillations, so different scientists started coming up with their own models and proposals of how such systems would interact.

One such scientist was Julian Hirniak, who in 1910 proposed a theoretical model for oscillations in cyclical reactions (Zhabotinsky, 1991). Hirniak's model can be summarised by the diagram in Figure. 4, and comprised three compounds which transformed into each other in two reaction cycles: one clockwise and one counterclockwise.

This system, too, can be written as a set of ODEs using the law of mass action. With six reactions total, each x_i participates in four of them in total. The ODE system has the

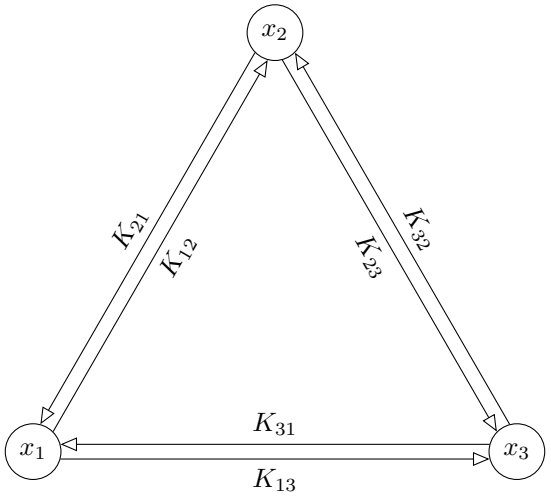


Figure 4: Hirniak's cyclical model for chemical oscillations. x_1, x_2 and x_3 represent concentrations of three different chemical compounds that react with each other with reaction rates given by the K_{ij} constants according to the arrows in the diagram.

form

$$\begin{aligned}
\frac{dx_1}{dt} &= -(k_{12} + k_{13})x_1 + k_{21}x_2 + k_{31}x_3, \\
\frac{dx_2}{dt} &= k_{12}x_1 - (k_{21} + k_{23})x_2 + k_{32}x_3, \\
\frac{dx_3}{dt} &= k_{13}x_1 + k_{23}x_2 - (k_{31} + k_{32})x_3.
\end{aligned}$$

Exercise 7:
Go through each of the reactions and recreate the ODEs yourself one by one. This is a good exercise to practice your knowledge of the law of mass action.

As before, one common assumption to make in a system of this form is that the sum of the concentrations is a constant value, x_T, such that

$$x_1 + x_2 + x_3 = x_T. \tag{32}$$

This constraint allows us to rewrite $x_3 = x_T - x_1 - x_2$ and reduce the system to a set of 2 ODEs given by

$$\begin{aligned}
\frac{dx_1}{dt} &= -(k_{12} + k_{13} + k_{31})x_1 + (k_{21} - k_{31})x_2 + k_{31}x_T, \\
\frac{dx_2}{dt} &= (k_{12} - k_{32})x_1 - (k_{21} + k_{23} + k_{32})x_2 + k_{32}x_T.
\end{aligned}$$

This system, however, cannot exhibit oscillations. Which is a little bit of a problem.

Exercise 8:

Find the coordinates of the steady state and calculate its stability matrix J. As a reminder, the corresponding characteristic polynomial has the form

$$\lambda^2 - T\lambda + \Delta = 0 \tag{33}$$

where T and Δ represent the trace and determinant, respectively. How does the stability result prevent the existence of *sustained* oscillations?

There is but one glimmer of hope yet, the system could still exhibit oscillations on the way to equilibrium (also known as *dampened* oscillations). This is where the *Principle of detailed balance* comes into play. This principle, introduced by the Austrian chemist Rudolf Wegscheider in 1901, states 'For each cycle oriented or an elemental process, the product of the kinetic constants is equal to the product of the kinetic constants of the inverse process' (Gorban and Yablonsky, 2011). In plain English this translates to

$$K_{12}K_{23}K_{31} = K_{13}K_{32}K_{21}.$$

Thus, the concentration of each and every compound must stay still during the reaction, and the cycles must constantly 'cancel' each other out. The reaction might still occur, but the three compounds must remain at the same ratios they were when the reaction started. This restriction is damning, and essentially restricts Hirniak's system from exhibiting any type of oscillatory behaviour.

Exercise 9:

Show that even dampened oscillations are not possible in the system by looking at the values of the discriminant given by

$$D = \sqrt{T^2 - 4\Delta}. \tag{34}$$

Hint: without loss of generality, you can assume that $k_{12} = \alpha k_{21}$ and $k_{23} = \beta k_{32}$ for some α and β.

The principle detailed balance is made in accordance to the *Second Law of Thermodynamics* (important stuff). The system must tend to equilibrium, after all (as all things in nature do). This conclusion made such an impact in the scientific community of the time that it lead to the belief that this principle must apply to every closed (i.e., one which can operate without constantly adding or removing compounds) chemical reaction. Chemical oscillations were completely and utterly out of the picture, and in doing so the field of study was sent into a 'dark age' (Zhabotinsky's words, not mine) (Zhabotinsky, 1991).

Searching for a chemical oscillator

The Brussels group

This does not mean that the interest in chemical oscillators disappeared completely during the era (but it does mean it became an underground subject). Ilya Prigogine was part of a group of researchers based in Brussels, Belgium. Ilya was interested in chemical oscillators, fighting an uphill battle against the rest of the field, but to do so he and the Brussels group needed to solve Hirniak's paradox. They argued that even in the system as a whole needed to tend towards equilibrium, there was nothing stopping the concentration of **intermediate compounds** (i.e., not initial reactants nor final products) from oscillating. Such oscillations were inherently transient, but did not violate the fundamental principles of the universe (Epstein and Pojman, 1998).

This observation about 'hidden intermediate compounds' was made in the 1950s from a completely theoretical point of view. It would take good decade and a half still for Prigogine and his group to present a chemical model that could exhibit these transient oscillations. In 1968 the Brussels group unveiled their crowning achievement, a model that is still studied to this day: the... (drumroll, please) **Brusselator** (Prigogine and Lefever, 1968). This system is described by the equations

$$\mathrm{A} \xrightarrow{k_1} \mathrm{X}, \tag{35a}$$

$$\mathrm{B} + \mathrm{X} \xrightarrow{k_2} \mathrm{Y} + \mathrm{D}, \tag{35b}$$

$$2\mathrm{X} + \mathrm{Y} \xrightarrow{k_3} 3\mathrm{X}, \tag{35c}$$

$$\mathrm{X} \xrightarrow{k_4} \mathrm{E}. \tag{35d}$$

Similar to Lotka's model, the only species of interest are X and Y, and the third reaction is autocatalytic on X. The Brussels group knew that such reactions were necessarily for oscillations to appear (partially thanks to Lotka). The 'net reaction' of the model can be obtained by adding up all the reaction, which results in

$$\mathrm{A} + \mathrm{B} \longrightarrow \mathrm{D} + \mathrm{E}.$$

And, like a magician making a coin disappear, the intermediate reactants X and Y are gone. The system does tend to equilibrium, A and B keep reacting with each other until completely transformed into D and E, but he also showed that such a system could produce oscillations! It was a completely theoretical system (at the time), but it solved Hirniak's paradox. Even then there was still a lot of pushback from the experimental groups at the time, would weren't going to let half a century of beliefs go without strong experimental evidence (and who can blame them really).

The work of the Brussels group did give a lot of hints, a 'laundry list' of requirements for an oscillating chemical reaction:

1. An oscillation could exist in a transient form, far from thermodynamic equilibrium.
2. The oscillating compounds must be intermediate reactants.
3. There has to be some sort of autocatalytic reaction involved.
4. The reaction can be maintained over long periods of time by introducing more A into the system.

The Brussels group never really found such a reaction on their own. On the other side of the iron curtain, in the heart of the Soviet Union, however, the search for a chemical oscillator was proving more fructiferous.

Mathematical analysis

The Brusselator as an ODE system

Before we head into the final chapter of our story, it is pertinent that we study the oscillations generated by the Brusselator. We will build on top of the phase plane analysis done in the Lotka system. Once again, we can use the Law of Mass Action to translate the system given by (35) into a set of ODEs.

The associated equations are

$$\frac{d\mathrm{X}}{dt} = k_1\mathrm{A} - k_2\mathrm{BX} + k_3\mathrm{X}^2\mathrm{Y} - k_4\mathrm{X}, \tag{36a}$$

$$\frac{d\mathrm{Y}}{dt} = k_2\mathrm{BX} - k_3\mathrm{X}^2\mathrm{Y}. \tag{36b}$$

All the k's are positive, of course. One topic I have neglected to mention so far are *units*. This model is meant to all represent real quantities are concentrations, so it is very important that both variables and reactions have the correct units. If you are asking 'why' in your head, that's a very 'mathematical' question to ask.

The reason I bring the topic of dimensional variables and constants for a very simple reason: I want to get rid of them. **Nondimensionalisation** is the process of reformulating a model in terms of dimensionless variables (Segel and Edelstein-Keshet, 2013). This is perhaps not ideal in terms of using system (36) to help explain experimental data (the experimentalists *hate* it), but it does have a few benefits (Segel and Edelstein-Keshet, 2013; Keener and Sneyd, 2010):

- It can reveal inconsistencies in formulation (i.e., adding together different variables or terms that have inconsistent units).
- It can reduce the number of free parameters, either by revealing dependencies between them or by highlighting ones that have a negligible effect on the model.
- It may reveal that some variables evolve way *faster* or *slower* than the other ones, which can results in reducing the dimension of the model (more on this later).

Notice that these are all 'maybe', there is no *a priori* guarantee that this will happen. Let's apply this concept to the Brusselator.

First, a bit of *mise in place*. What are the components of the system?

- Variables: X, Y (2).
- Parameters: k_1, k_2, k_3,k_4, A, B (6).

Note that, for example, A and B can be reformulated as

$$\alpha = k_1 A \qquad\qquad \beta = k_2 B$$

which allows us to 'absorb' two of the parameters, since they do not appear anywhere else in the equations.

Additionally, we can introduce a new set of variables

$$x = \frac{X}{x_0} \qquad y = \frac{Y}{y_0} \qquad \tau = \frac{t}{T_0}. \tag{37}$$

In this case, I will cheat a little bit by choosing $x_0 = y_0 = 1$. There are other possible choices, but this one will suffice for now. Using the chain rule, we can rewrite the system using this variable into

$$\frac{dx}{d\tau} = (T_0)\,\frac{dX}{dt} = (T_0)(k_1\mathrm{A} - k_2\mathrm{B}x + k_3x^2y - k_4x), \tag{38a}$$

$$\frac{dx}{d\tau} = (T_0)\,\frac{dY}{dt} = (T_0)\,(k_2\mathrm{B}x - k_3x^2y). \tag{38b}$$

Four possible choices emerge for T_0

$$T_0 = \frac{1}{k_1} \qquad T_0 = \frac{1}{k_2} \qquad T_0 = \frac{1}{k_3} \qquad T_0 = \frac{1}{k_4}. \tag{39}$$

Lets set the first aside for now and focus on the last two. either $T_0 = k_3$ and we get rid of k_3 or $T_0 = k_4$, and we eliminate k_4. Let us make the first choice to simplify the quadratic term and define

$$k = \frac{k_4}{k_3}, \tag{40}$$

to get the more 'classic' ODE version of the Brusselator, given by

$$\frac{dx}{dt} = \alpha - \beta x + x^2y - kx, \tag{41a}$$

$$\frac{dy}{dt} = \beta x - x^2y \tag{41b}$$

Were the dropped the bar notation for convenience (just do not tell the pure mathematicians). There is no unique way to do this process, and the best method often depends on what you want to do with the model. Nondimensionalisation is as much an art form as anything.

Exercise 10:

Repeat the nondimensionalisation process with the dimerisation reaction model given by

$$\mathrm{A} + \mathrm{A} \underset{k_-}{\overset{k_+}{\rightleftarrows}} \mathrm{C}. \tag{42}$$

Phase plane analysis

Let us call the nullclines N_x and N_y. In this system they are given by

$$N_x := \left\{ y = \frac{(\beta+1)x - 1}{\alpha x^2} \right\}, \tag{43}$$

$$N_y := \left\{ y = \frac{\beta}{\alpha x} \right\}. \tag{44}$$

Two important states are the origin and $(x, y) = (\alpha/k, k\beta/\alpha)$. Let us call this steady state (x_0, y_0). The stability matrix (Jacobian) is generically given by

$$J = \begin{pmatrix} -\beta + 2xy - k & x^2 \\ \beta - 2xy & -x^2 \end{pmatrix} \tag{45}$$

At the special steady state

$$J = \begin{pmatrix} -\beta + 2x_0y_0 - k & x_0^2 \\ \beta - 2x_0y_0 & -x_0^2 \end{pmatrix} = \begin{pmatrix} \beta - k & \alpha^2/k^2 \\ -\beta & -\alpha^2/k^2 \end{pmatrix}. \tag{46}$$

The characteristic polynomial is given by

$$\lambda^2 + \left(\frac{\alpha^2}{k^2} + k - \beta\right)\lambda + \left(\frac{\alpha}{k}\right)^2 = 0.$$

Exercise 11:

Time to bring out the pen and paper. Calculate the explicit form of the eigenvalues λ_1 and λ_2.

The determinant $(\alpha/K)^2$ is always positive, thus both eigenvalues will share signs. The steady state can never be a saddle point.

Note that at $\beta = 0$ both eigenvalues have positive real part, and as the value of B increases past $(\alpha/k)^2 + k$ the stability of the steady state changes. The fixed point loses its stability and the trajectories that once all converged into it suddenly leave it instead (leave **where**?, you may ask).

This represents a fundamental change in the qualitative behaviour of the system, as the shape of the trajectories near the steady state is radically different as β crosses the value $(\alpha/k)^2 + k$. We cal these behavioural transitions **bifurcations**.

When β crosses this **bifurcation value**, the eigenvalues become purely imaginary numbers. They also move across the imaginary axis as B keeps increasing, instead of staying stuck at it. This particular transition is called a **Hopf bifurcation**, and its consequences are summarised by the following theorem.

Hopf bifurcation: Here's the hand-waving statement of the theorem. Suppose that a steady state changes from a spiral in to a spiral out, as some parameter β crosses some critical value, 0 say. (It could be any value, but we let it be zero without loss of generality.) Then, for some range of values of β, either below or above 0, there is a limit cycle. The amplitude of this limit cycle tends to zero as $\beta \longrightarrow 0$, but its period doesn't.

Slightly more formally, suppose that a dynamical system has a steady state $\mathbf{x}_0$, and around this steady state the linearisation has a pair of complex conjugate eigenvalues $\omega_0(\beta) \pm i\omega_1(\beta)$. If

- $\omega_0(0) = 0$ (eigenvalues are pure imaginary at the critical point)
- $\left.\frac{d\omega_0}{d\beta}\right|_{\beta=0} \neq 0$ (the eigenvalues cross over the imaginary axis)
- $\omega_1(0) \neq 0$ (so that there is still an imaginary part at the bifurcation point)

then there exists a one-parameter family of limit cycles, parameterised by β, existing

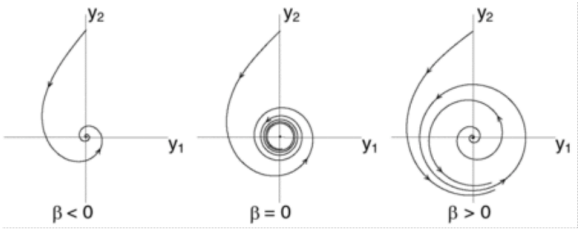


Figure 5: This first case is called a supercritical Hopf bifurcation (i.e., super means the limit cycles appear **after** the bifurcation point, as the steady state goes from stable to unstable). The limit cycles are stable.

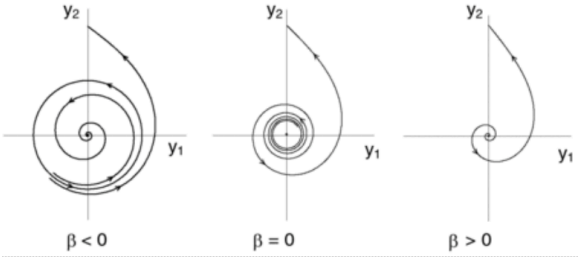


Figure 6: The second case is called a subcritical Hopf bifurcation, as the limit cycles appear **before** the bifurcation point (again, as the steady state goes from stable to unstable). The limit cycles are now unstable.

either for some neighbourhood $\beta > 0$ or for some neighbourhood $\beta < 0$.

Figs. 5 and 6 show diagrams of the two possibilities (the limit cycle occurring either for $\beta > 0$ or for $\beta < 0$).

Indeed, our discovery of a Hopf bifurcation guarantees the existence of a **limit cycle**. A limit cycle is an **attractor**, an object similar to a steady state, but this time it is a closed curve instead of a singular point. In this case, the loss of stability of the steady state guarantees that the limit cycle will be stable. The trajectories 'nearby' will be pulled towards it over time, and they will 'copy' its shape. (Fun fact: as with steady states, trajectories never really touch the limit cycle, they are merely infinitesimally away from it). But, the more important part, is that we have found an oscillator within this system!

Exercise 12:

Consider the Van del Pol oscillator on its two-dimensional form

$$\frac{dx}{dt} = y, \tag{47}$$

$$\frac{dy}{dt} = \mu(1 - x^2)y - x. \tag{48}$$

This system contains a classic example of a Hopf bifurcation for the parameter μ.

- Find the bifurcation value.
- Calculate the form of the eigenvalues.
- Figure out if the bifurcation is sub- or supercritical.

Since the bifurcation value depends on A, the bifurcation condition $\beta = (\alpha/k)^2 + k$ actually defines a curve in the α/β plane. It is very tempting to conclude that this curve bisects the plane into two regions: one where oscillations occur, the other were they don't. Actually, that's a very strong statement, which doesn't follow from the Hopf bifurcation theorem, which applies only locally to the steady state. That is, the HB theorem guarantees the existence of oscillations only in some neighbourhood (possibly very small) of the

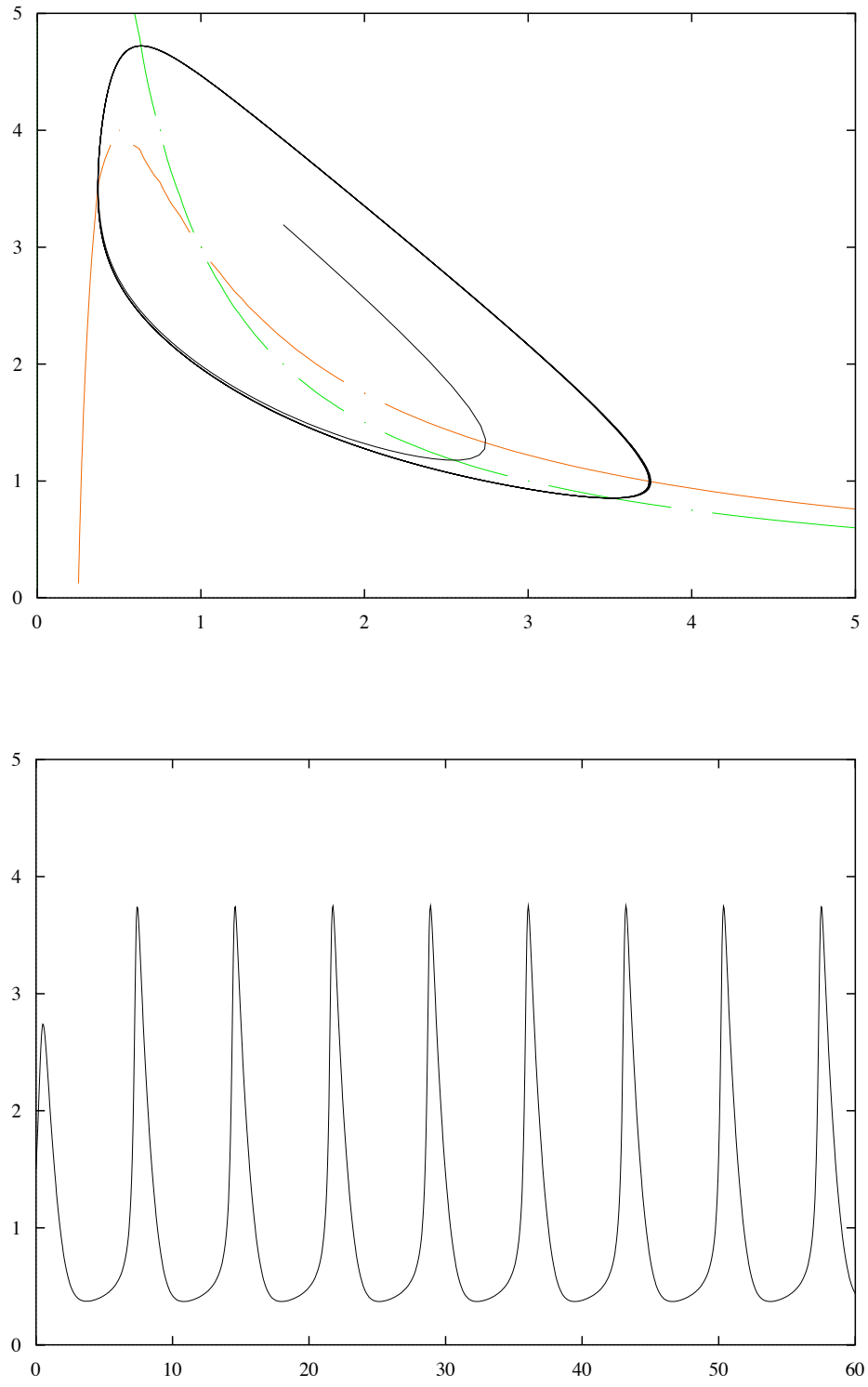


Figure 7: Top: phase plane of the Brusselator, with a solution. $A = k = 1$, $B = 3$. Bottom: Time series of the model. Axes are unlabelled because I'm lazy and can't be bothered, but the right-hand panel is X.

bifurcation curve. In addition, the HB theorem doesn't tell us whether the limit cycles occur below or above the HB curve.

To secure the existence of this limit cycle on a bigger region of the parameter space, we need to resort to another very important theorem from dynamical systems.

The Poincaré-Bendixson Theorem

Given a differentiable real dynamical system defined on an open subset of the plane, then every non-empty compact ω-limit set of an orbit, which contains only finitely many fixed points, is either

- a fixed point,
- a periodic orbit, or
- a connected set composed of a finite number of fixed points together with homoclinic and heteroclinic orbits connecting these.

The ω-limit set of an orbit is essentially the set of points reached by the orbit as $t \to \infty$ (although it has a more formal definition which you can learn about in MATH 761). For example, if the orbit tends to a stable node, then the ω-limit set is the node itself. If the orbit tends to a stable limit cycle, then the ω-limit set is the limit cycle.

How does this help? Well, suppose we can find a region in the plane from which no trajectories can leave (and into which some trajectories enter). If this **trapping region** contains no fixed points (or if the fixed points are completely unstable, like in our case), then every trajectory entering this region must tend to a stable limit cycle. There is no other option.

Every trajectory entering this region can't just stop (as there are no fixed points) and it can't intersect itself (or any other trajectories) as the solutions are unique, so what can it do? All it can do is wind round, getting closer and closer to a limit cycle. This result is (unfortunately) confined to the two-dimensional case, as in higher dimensions there is so much more 'space' for the trajectories to move around and wind in very *chaotic* ways. Fortunately, we don't have to worry about that in our case.

The trapping region

Creating a trapping region is a very artisanal process. To make one we need to find boundaries where the flow always points towards the inside of the region. There is no pre-determined way of creating these regions, and in a lot of cases it involves a lot of trial and error.

For the Brusselator, we can make a trapping region by looking at

- L_1. The line $x = 0$. Easy to see that $\dot{x} \geq 0$ all along this line and thus the solutions are pointing in everywhere. The only possible difficulty is at $(x, y) = (0, 0)$, but that's homework.
- L_2. The line $y = 0$. Clearly $\dot{y} > 0$ along the axis, as is indeed required for the model to be acceptable at all.
- L_3. The line with slope -1 cutting across. Along this line we have $(\dot{x}, \dot{y}) \cdot (-1, -1) = kx - \alpha$ which is greater than 0 as long as $x > \alpha/k$, the value at the fixed point. So as long as don't extend this line past the x value of the fixed point we'll be OK.

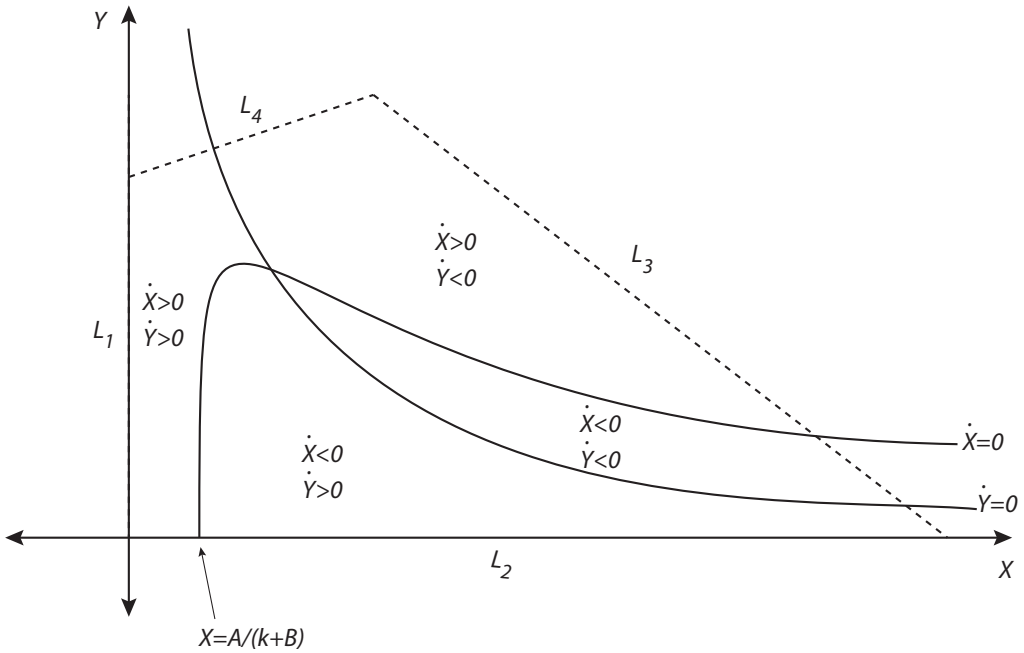


Figure 8: The phase plane for the Brusselator.

- L_4. The line $y = x + c$ for some c. We just need to work out how big c has to be. The inward normal to L_4 is $(1, -1)$. We have

$$(\dot{x}, \dot{y}) \cdot (1, -1) = (2y)x^2 - (2\beta + k)x + a,$$

  which is everywhere greater than zero as long as it has no real roots. But this quadratic has no real roots if $y > \frac{(2\beta+k)^2}{8\alpha}$ so all we need is $c > \frac{(2\beta+k)^2}{8\alpha}$. I don't think this is a necessary condition (I haven't checked) but it's certainly sufficient. In a very scientific way, just make y big enough.

We're almost done. We've now found a region that has trajectories always entering it. However, it still contains a fixed point. So, we draw a small circle around the unstable fixed point, and we know that all solutions must cross from the inside to the outside of this circle (since this is what it means for the fixed point to be an unstable node). Fig. 9 gives a picture of the trapping region and the box.

Bingo. We're done. We've now found a region (the trapping region minus the circle) that contains no fixed points, but has solutions always entering it. It must contain a stable limit cycle, which is what we wanted in the first place. A typical solution in the phase plane and as a function of time is shown in Fig. 7.

Exercise 13:

Try to construct a different trapping region for the Brusselator. Hint; on the left and right sides use vertical lines, and put a horizontal line across the top.

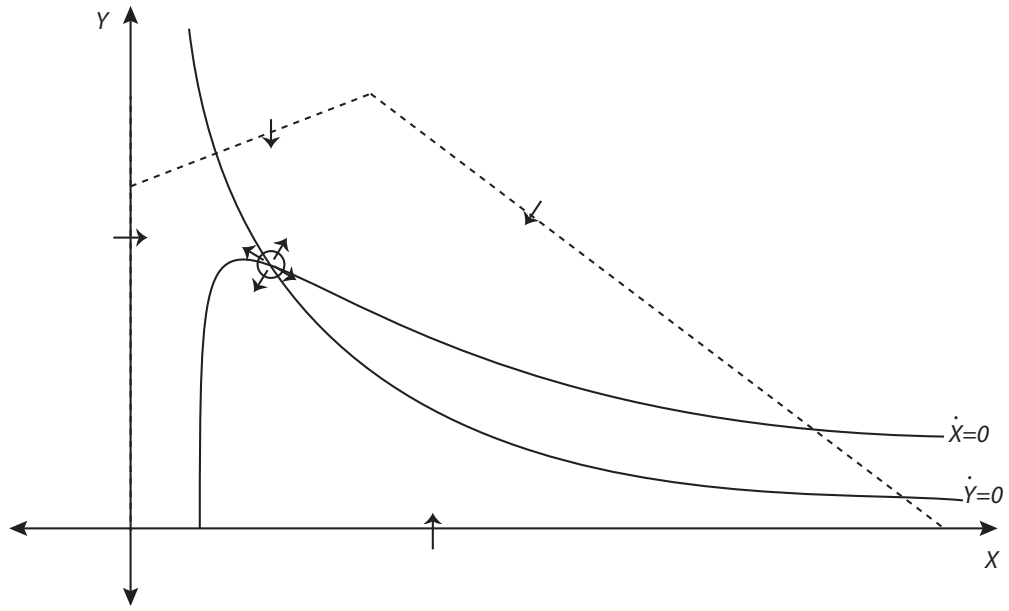


Figure 9: The trapping region for the Brusselator.

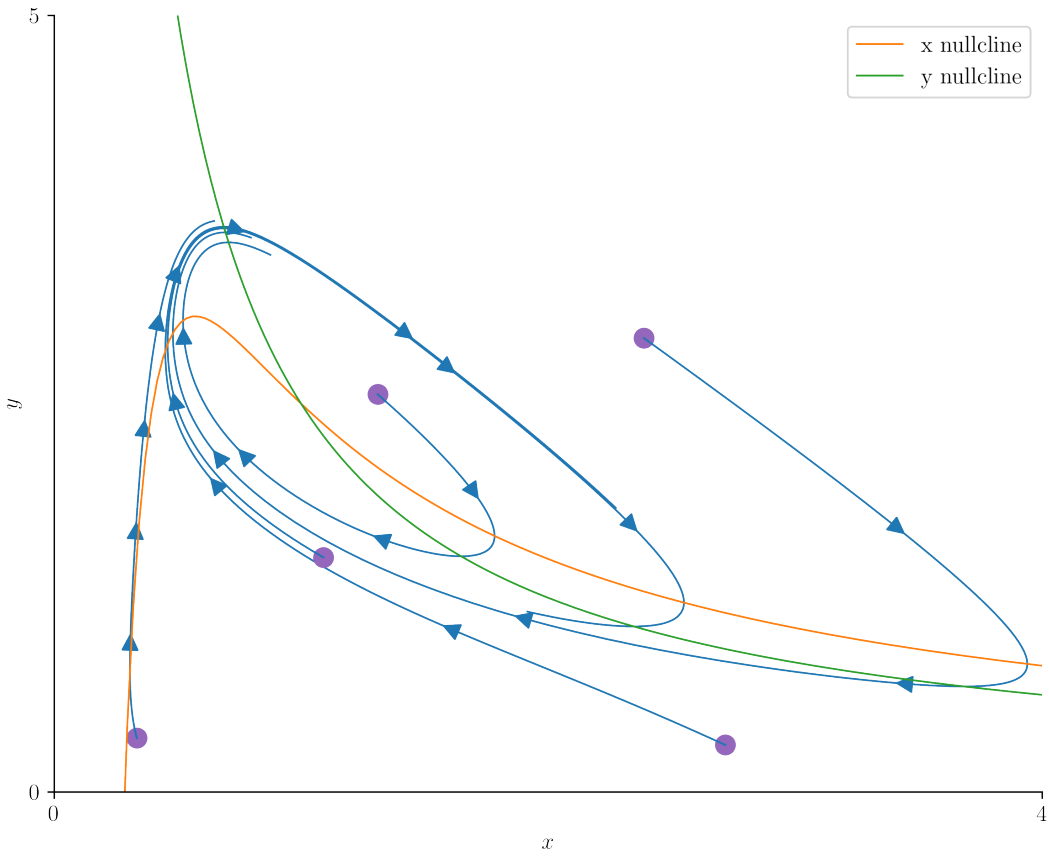


Figure 10: The phase plane for the Brusselator along with some trajectories starting at the purple dots.

The Belousov-Zhabotinsky reaction

Boris Belousov

Boris Pavlovich Belousov was a Soviet chemist and biophysicist who was working in the Soviet Union around the middle of the XX century. His research centred around the *Krebs cycle*, the series of reaction that allow us to obtain energy from food. The reactions involved are as far away from trivial as possible, reason why Belousov was trying to find a simple inorganic reaction that he could study as a proxy of the real cycle.

Belousov knew that citric acid ($C_6H_8O_7$) was a fundamental part of the real cycle. In one of his experiments in 1951, he decided to mix citric acid with sodium bromate ($NaBrO_3^-$) and a cerium compound to create a facsimile of the cycle (Tyson, 1994). Once he mixed all the ingredients in a flask, he noticed that the resulting colorless solution changed to a pallid yellow colour after a minute.

And then changed back to colorless after another minute.

And then regained the yellow tint, after roughly sixty seconds.

This behaviour continued roughly for an hour, after which the solution stabilised (Zhabotinsky, 1991; Tyson, 1994; Epstein and Showalter, 1996). Belousov was pretty confident that there was no external influence acting on this flask, and it seemed that the concentration of some undetermined compound kept periodically changing on this flask.

Indeed, Belousov had found a prime example of an oscillating inorganic chemical reaction. Not only that, the reaction seemed relatively simple to recreate, it also used relatively cheap and available ingredients. All of these factors seemed to position this bromate-citric acid reaction as the perfect candidate to break the half-century paradigm on oscillatory reaction. Belousov certainly thought so, since very shortly thereafter he rushed to publish his results.

And this is where we get to the crux of our story. Said simply, yes, Belousov had found an oscillatory chemical reaction, there are no if and buts about it. In fact, he had not even been the first one: one year after the publication of Lotka's model, William Bray (a Canadian chemist) had found a different oscillatory reaction. But, as happened with Bray almost *40 years before*, Belousov faced enormous pressure when trying to publish his results. Biologist Arthur Winfree notes on his historical review (Winfree, 1984) that the reviewers assigned to Belousov were so sure that the Laws of Thermodynamics forbade the existence of simple chemical oscillators that they conditioned their approval of his manuscript on Belousov *proving that the entire current understanding of chemical oscillators was wrong.* It suffices to way that this was *wildly* out of scope for what Belousov was trying to publish.

His work remained unpublished for almost a decade, until he managed to sneak it in a journal for medicinal radiation (which was published by the same institute he worked

in). The manuscript was not even published on its entirety, but it would eventually be recovered decades later (Zhabotinsky, 1991). Contained in it were proofs or hints of many advancements that would be made in laboratories all over the world in the years after his discovery.

Anatoly Zhabotinsky

About a decade after Belousov's discovery, a postgraduate student by the name of Anatoly Zhabotinsky would come across his recipe while studying glycolysis[2]. Zhabotinsky noted that the change of colour in the reaction was due to cerium ions changing from their Ce^{3+} (colorless) configuration to Ce^{4+} (pale yellow). He changed some ingredients, mostly importantly swapping cerium for ferroin (an iron-based compound) which changed the colour gradient from transparent-white into a much more contrasting blue-red. The channel *NileRed* on YouTube has a particularly good recreation of this reaction.

Zhabotinsky also came up with a possible mechanistic explanation behind the oscillations, which is presented in Figure 11. Notice that bromine based compounds make up the initial reactants and the final products. Cerium, crucially, is *neither*. As the Brussels group predicted, the oscillating compound was an intermediary, hidden in the middle of the reactions. You will remember (because of course you have been paying attention), that the other big prerequisite identified by the Brussels group was the presence of an *autocatalyst*. Zhabotinsky represents it as the big 'X' in the middle of the diagram, not because he loved the suspense, but because he could not identify it at the time.

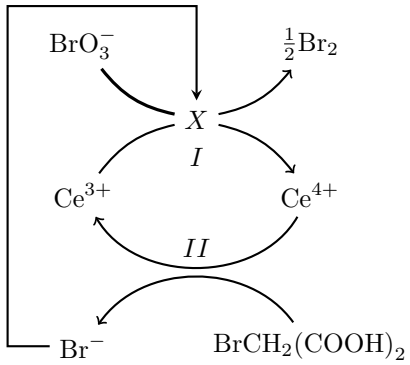


Figure 11: Zhabotinsky first attempt to describe the BZ reaction. Zhabotinsky (1991).

Zhabotinsky went to great lengths to prove that the reaction represented a genuine example of an oscillator, upscaling the reaction 1000-fold to prove that there were no small impurities that could be blamed for the behaviour. He even used Lotka's model as the basis for a theoretical explanation behind the existence of the oscillation, breaking the great barrier that stopped Belousov from publishing (Zhabotinsky, 1991).

The FKN mechanism

Moving forward to the 1960s, Zhabotinsky and his group were offered a rare and unusual opportunity to leave the Soviet Union and present their research at a conference in Prague

[2]Apparently, he was originally trying to study a different reaction, but it used some really expensive compounds, and he was handed the recipe as a 'cheaper' alternative to test his theories on.

in 1968. There they met the Brussels group. Suffice to say they were thrilled to hear about his work, as was an American professor named Robert Mazo.

Mazo brought the information about the BZ reaction back to the University of Oregon, where he worked and presented it in a colloquium (back when you couldn't just Zoom this kind of information in your second screen and vaguely pay attention to it). Richard Noyes and Richard Field, kineticists, were so intrigued by this information that they decided to study it alongside professor Endre Kőrös (who was part of the Brussels group and was on sabbatical at Oregon). After three years of research they produced a full mechanistic explanation behind the BZ reaction, which is known as the FKN mechanism. After three decades and the work of researchers all across the world, the BZ reaction was finally described in a set of simple, understandable set of reactions given by

$$\mathrm{HOBr} + \mathrm{Br}^- + \mathrm{H}^+ \rightleftarrows \mathrm{Br}_2 + \mathrm{H}_2\mathrm{O}, \tag{R1}$$

$$\mathrm{HBrO}_2 + \mathrm{Br}^- + \mathrm{H}^+ \to 2\mathrm{HOBr}, \tag{R2}$$

$$\mathrm{BrO}_3^- + \mathrm{Br}^- + 2\mathrm{H}^+ \to \mathrm{HBrO}_2 + \mathrm{HOBr}, \tag{R3}$$

$$2\mathrm{HBrO}_2 \to \mathrm{BrO}_3^- + \mathrm{HOBr} + \mathrm{H}^+, \tag{R4}$$

$$\mathrm{BrO}_3^- + \mathrm{HBrO}_2 + \mathrm{H}^+ \rightleftarrows 2\mathrm{BrO}_2 + \mathrm{H}_2\mathrm{O}, \tag{R5}$$

$$\mathrm{BrO}_2 + \mathrm{Ce}^{3+} + \mathrm{H}^+ \rightleftarrows \mathrm{HBrO}_2 + \mathrm{Ce}^{4+}, \tag{R6}$$

$$\mathrm{BrO}_2 + \mathrm{Ce}^{4+} + \mathrm{H}_2\mathrm{O} \to \mathrm{BrO}_3^- + \mathrm{Ce}^{3+} + 2\mathrm{H}^+, \tag{R7}$$

$$\mathrm{Br}_2 + \mathrm{CH}_2(\mathrm{COOH})_2 \to \mathrm{BrCH}_2(\mathrm{COOH})_2 + \mathrm{Br}^- + \mathrm{H}^+, \tag{R8}$$

$$6\mathrm{Ce}^{4+} + \mathrm{CH}_2(\mathrm{COOH})_2 + 2\mathrm{H}_2\mathrm{O} \to 6\mathrm{Ce}^{3+} + \mathrm{HCOOH} + 2\mathrm{CO}_2 + 6\mathrm{H}^+, \tag{R9}$$

$$4\mathrm{Ce}^{4+} + \mathrm{BrCH}_2(\mathrm{COOH})_2 + 2\mathrm{H}_2\mathrm{O} \to \mathrm{Br}^- + 4\mathrm{Ce}^{3+} + \mathrm{HCOOH} + 2\mathrm{CO}_2 + 5\mathrm{H}^+. \tag{R10}$$

Ok, maybe I exaggerated on the 'simple' part. Fear not, for they also identified that the 'core' of the reaction reduced down to only three chemicals: cerium (Ce^{4+}), bromous acid (HBrO_2) and bromine (Br^-). They wanted to extract this core, the part that actually enabled the oscillations, which they summarised in a 5 equation model known as (drumroll, once again)... the **Oregonator**.

$$A + Y \to X + P, \tag{50a}$$

$$X + Y \to 2P, \tag{50b}$$

$$A + X \to 2X + 2Z, \tag{50c}$$

$$2X \to A + P, \tag{50d}$$

$$B + Z \to \frac{1}{2} fY. \tag{50e}$$

Here $\mathrm{X} = [\mathrm{HBrO}_2]$, $\mathrm{Y} = [\mathrm{Br}^-]$ and $\mathrm{Z} = [\mathrm{Ce}^{4+}]$. The reaction rates are given by $k_1, \cdots, k_5$. A, B and P are the usual necessary compounds for the reaction to work. f is a special scaling parameter which we will discuss later. Notice that they also identified the autocatalyst: X (following the third reaction). As before, we can use the Law of Mass Action to transform

this system into a set of ODEs, which has the form:

$$\frac{dX}{dT} = k_1 AY - k_2 XY + k_3 BX - 2k_4 X^2, \tag{51a}$$
$$\frac{dY}{dT} = -k_1 AY - k_2 XY + \frac{f}{2} k_5 Z, \tag{51b}$$
$$\frac{dZ}{dT} = 2k_3 BX + k_5 Z, \tag{51c}$$

Does this system actually exhibit oscillatory solutions? Of course! I wouldn't have gone through all the trouble if it did not work as expected. But that does not mean we get to skip the mathematical analysis. *Au contraire*, we are going to need even more profound mathematics before being able to describe *how* these oscillations happen.

Mathematical analysis

Before we jump into the analysis, it is time to remember that the Poincaré-Bendixson theorem (among other things) does not work in the three-dimensional case. What to do about it? Well, we could this nondimensionalisation and see if that helps us simplify the system.

Nondimensionalisation

Lets us do this one more time. Choosing the parameters for nondimensionalisation is less obvious in this system when compared to the Brusselator, due to the complexity of the equations.

In this case we will resort to the more 'generic' form of nondimensionalisation, which assumes we are going to rescale everything and seeing where that leads us. Consider the following dimensionless variables

$$u = \alpha X, \qquad v = \beta Y, \qquad w = \gamma Z, \qquad t = \delta T. \tag{52}$$

First step is to obtain the corresponding differential equations for u, v and w. In doing so, hopefully we can find good candidates for α, β, γ and δ. We? Sorry, I meant you.

Exercise 14:

Nondimensionalise the Oregonator. I will guide you through it, do not worry.

1. Find the differential equations for the dimensionless parameters. You will have to do a double substitution in each one to change both the state variable (i.e., u) and the time variable (t).
2. Reorder the equations so that each term has the form (parameter) * (variable(s)). In doing so you will see multiple candidates for α and β.
3. A very convenient choice to make for α (following Tyson† (1979)) is one that simplifies the quadratic term u^2 (i.e., makes the corresponding coefficient 1).
4. Choose β to in a way that simplifies the uv term.
5. Choose γ to simplify the w in the last equation.

6. Finally, pick δ to simplify the w equation.

7. You will end up with a very nasty fraction in the v terms. You can condense all the parameters involved into a new parameter

$$\mu = \frac{2k_1k_4}{k_2k_3}. \tag{53}$$

If you followed the steps correctly, you will end up at a system of the form

$$\epsilon\dot{u} = \mu v - uv + u - u^2, \tag{54a}$$
$$\epsilon'\dot{v} = -\mu v - uv + fw, \tag{54b}$$
$$\dot{w} = u - w. \tag{54c}$$

where

$$\epsilon = \frac{k_5B}{k_3A} \qquad\qquad \epsilon' = \frac{2k_4k_5B}{k_2k_3A}. \tag{55}$$

These are known as the **Field-Noyes** equations. These new ϵ and ϵ' parameters are usually referred to as **time-scales**, and they quantify how fast or slow u and v evolve relative to w (as a consequence of setting our frame of reference at w). We haven't talked about parameter values much, but according to the most common parameter set, these parameters have the values (Field and Noyes, 1974)

$$\epsilon \approx 1.4 \times 10^{-2}, \qquad \epsilon' \approx 3.7 \times 10^{-5}, \qquad \mu \approx 9.5 \times 10^{-5}. \tag{56}$$

This implies that, if the right-hand sides were all of roughly similar magnitude, u evolves around 100 times faster than w does, and v a thousand times faster than that. Imagine something moving 10,000 times faster than you. At that lightning pace, it might as well reach the finish line before you start moving. On that note, given these parameters values it is not out of the question to do an *approximation* and assume that

$$\epsilon' = 0 \tag{57}$$

(if we wanted to be more rigorous here, we would say that we are looking at the asymptotic behaviour of the system as $\epsilon' \to 0$). This removes the v equation and instead implies that v reaches its 'final state' before the rest of the system starts moving. This allows us to set v as a function of u and w via the restriction

$$-\mu v - uv + fw = 0. \tag{58}$$

In other words

$$v = \frac{fw}{u + \mu}. \tag{59}$$

We can substitute this back into the other two equations and obtain the two-dimensional version of the Oregonator.

$$\epsilon\dot{u} = u - u^2 - fw\frac{u - \mu}{u + \mu}, \tag{60a}$$
$$\dot{w} = u - w. \tag{60b}$$

Phase plane analysis

I will rush through the details here, after all this does not differ that much from the work we did in the Brusselator. The nullclines for this system are given by

$$N_u := \left\{ w = \frac{1}{f}\left(u(1-u) + 2\mu u \frac{1-u}{u-\mu} \right) \right\}, \tag{61}$$

$$N_w := \{ u = w \}. \tag{62}$$

As usual, the origin is a possible steady state, with the other two given by the solutions of the quadratic equation

$$u^2 - (1-\mu-f)u - \mu(f+1) = 0. \tag{63}$$

There are two possible roots, one negative (inconsequential) and one positive given by

$$u^* = w^* = \frac{1}{2}\left(1 - (\mu+f) + \sqrt{(\mu+f-1)^2 + 4\mu(f+1)} \right). \tag{64}$$

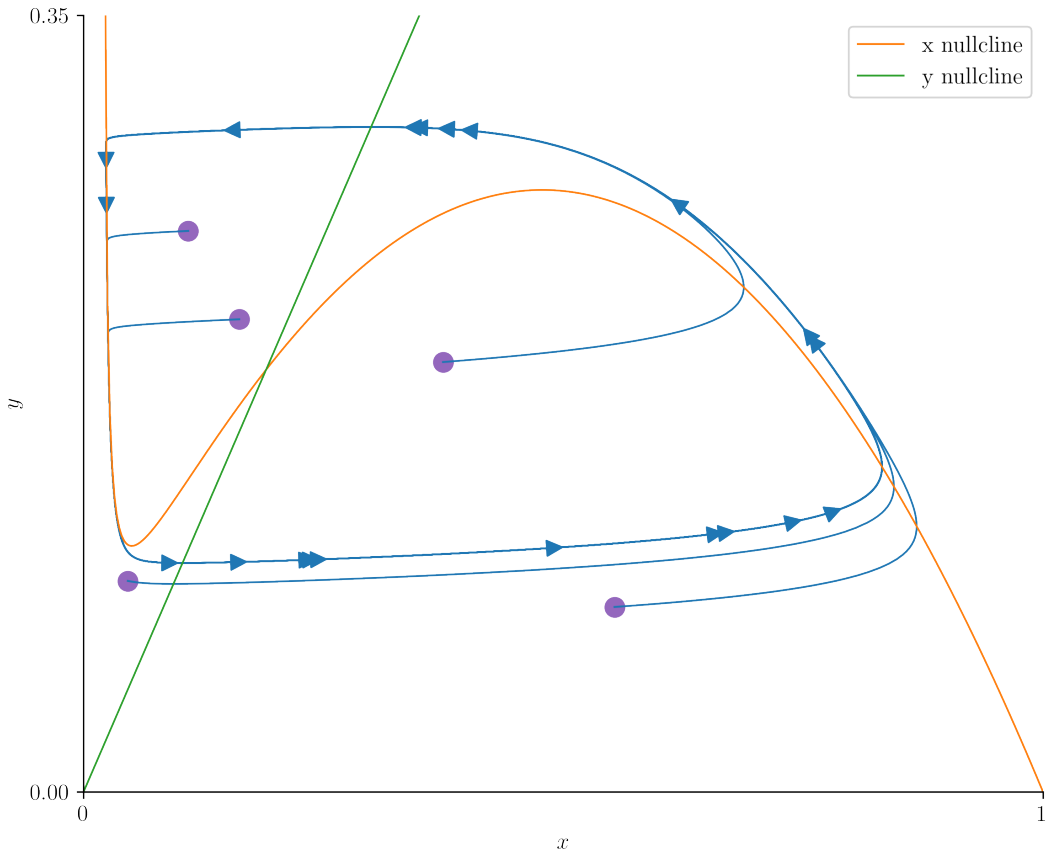


Figure 12: Phase plane of the simplified Oregonator. N_u is shown in blue and N_w in orange. The non-trivial steady state is shown in red. Four different trajectories start at the purple points and flow in the direction of the arrows. Parameter values are $f = 1$ and $\mu = 0.02$

To find the stability of the (u^*, w^*) steady state we look at the stability matrix given by

$$J(u,w) = \begin{pmatrix} \frac{1}{\epsilon}\left(1 - 2u - \frac{2\mu f w}{(u+\mu)^2}\right) & -\frac{1}{\epsilon} f\left(\frac{u-\mu}{u+\mu}\right) \\ 1 & -1 \end{pmatrix} \tag{65}$$

which, evaluated at the steady state, has the form

$$J(u,w) = \begin{pmatrix} \frac{1}{\epsilon}\alpha & -\frac{1}{\epsilon}\beta \\ 1 & -1 \end{pmatrix} \tag{66}$$

where

$$\alpha = 1 - 2u^* - \frac{2\mu f u^*}{(u^*+\mu)^2}, \qquad \beta = f\left(\frac{u^*-\mu}{u^*+\mu}\right). \tag{67}$$

The characteristic polynomial has the very convenient form

$$\lambda^2 + \left(1 - \frac{\alpha}{\epsilon}\right)\lambda + \left(\frac{\beta - \alpha}{\epsilon}\right) = 0. \tag{68}$$

At this point the only two parameters of importance in the system are μ and f. The sign of the determinant can be determined by substituting and simplifying the $\beta - \alpha$ expression, which (after a lot of algebra) results in

$$\beta - \alpha = \frac{(u^*)^2 + \mu u^* + 2\mu w^*}{u^* + \mu} > 0. \tag{69}$$

As before, this means the real part of both eigenvalues with always share sign. What about the trace?

- If $\alpha > \epsilon$ then the trace is positive, so are the eigenvalues and the steady state is unstable.
- If $\alpha < \epsilon$ then the trace is negative, so are the eigenvalues and the steady state is stable.

Which is the exact same setup needed for a Hopf bifurcation! If only it were that simple. Remember, f is the relevant parameter, not α. Have a look at the expression for α and remember that u^* in that expression follows equation (64). You can try to write out α as a function of f, I have done so in the past, but you will quickly realise that the expression is unwieldy (to say the least). Not to talk about writing the condition $\alpha = \epsilon$ in terms of f.

It is possible to get an approximation for the bifurcation values by abusing the fact that $\mu \approx 0$ and doing a lot of algebra. Tyson† (1979) has a brief overview of this process, but he does not include all the small details either. I will jump ahead of the computations for now and instead just say that, for $\epsilon \approx 0$, the system has two Hopf bifurcations for f at values

$$f_{H_1} \approx \frac{1}{2} \qquad \text{and} \qquad f_{H_2} \approx 1 + \sqrt{2}. \tag{70}$$

And that, for $f < f_{H_1}$ and $f > f_{H_2}$, the steady state is stable; on the other hand, if $f_{H_1} < f < f_{H_2}$ then the steady state loses stability. The system undergoes two *supercritical* Hopf bifurcations, and a limit cycle is guaranteed to appear nearby f_{H_1} and f_{H_2}. As you might recall, to ensure the existence of the stable limit cycle in the interval (f_{H_1}, f_{H_2}) we need to construct a trapping region. This will be left as an exercise, but I will give you some hints.

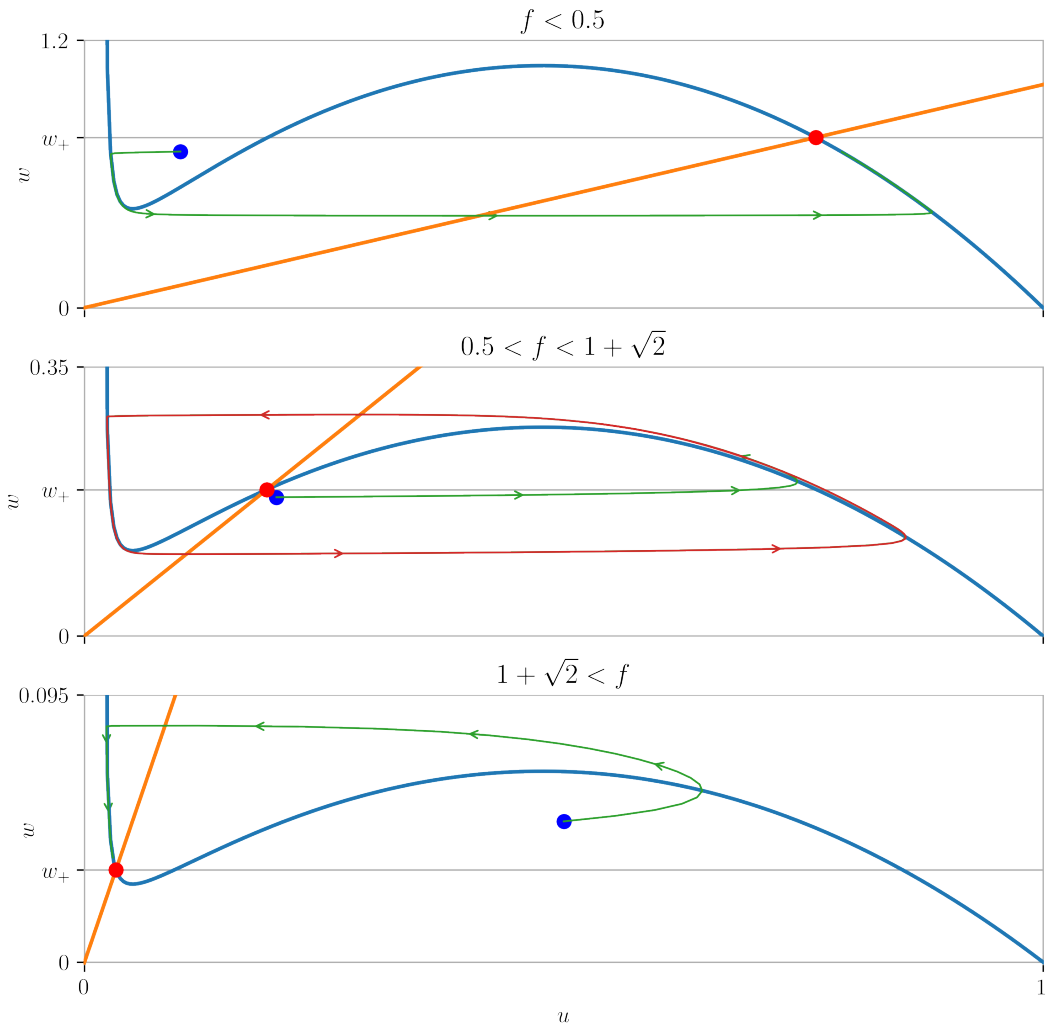


Figure 13: Phase plane of the simplified Oregonator model, for three different f values, corresponding to $f < f_{H_1}$, $f_{H_1} < f < f_{H_2}$ and $f_{H_2} < f$

Exercise 15:

Construct a trapping region for the simplified Oregonator, in the case where the steady state is unstable. Some pointers to keep in mind.

1. Remember that the region must not contain *any* fixed points. If you are going to include the origin, you must first prove that it is completely unstable (and if it is not, then you must find a workaround).

2. The simplest way to construct a trapping region for this system is to draw a rectangle. You must find all the corners to do so. I can tell you that (1,1) is a good candidate for a corner.

3. Remember to check that the flow at the edges of your trapping box points inwards

(you can do this by calculating the dot product).

4. As per a previous comment, both ϵ and μ are approximately (but not exactly!) 0.

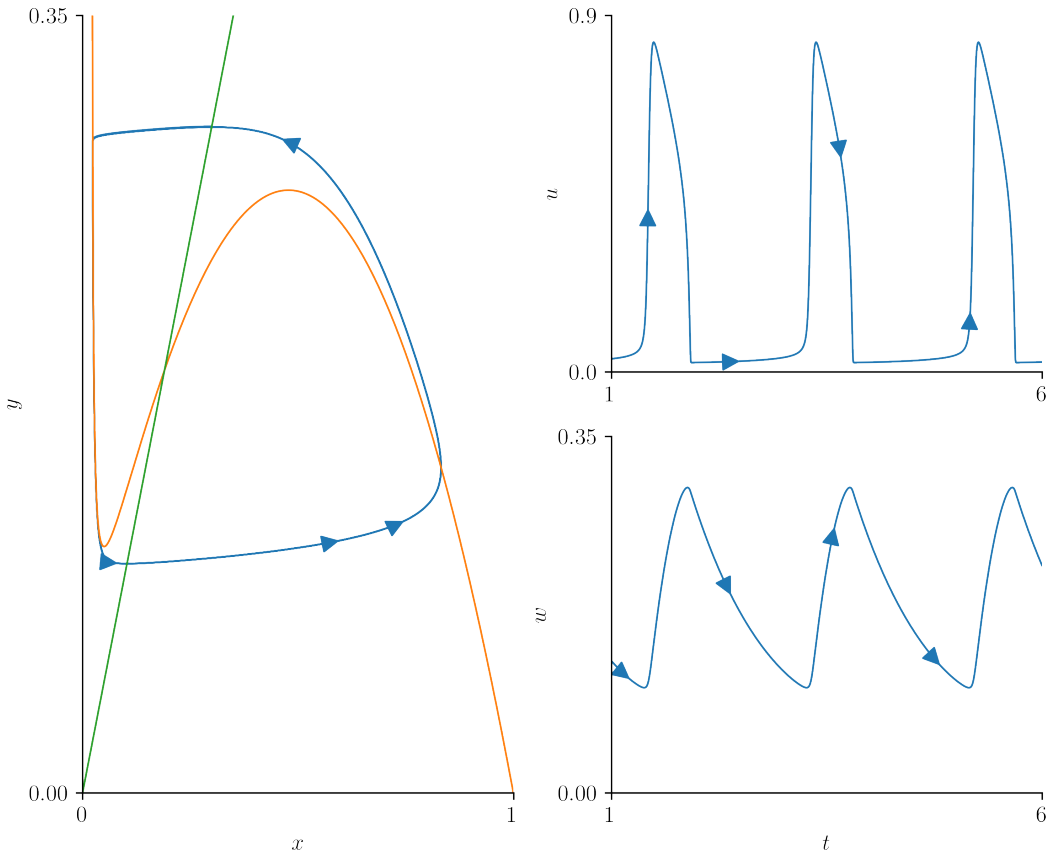


Figure 14: Phase plane of the simplified Oregonator, alongside a periodic orbit obtained for $f = 1$ and $\mu = 0.02$. The left panels show the time series for both variables.

With the trapping region in place we can be confident that we have done our due diligence and that the existence of the limit cycle is secured. Next up, here is a really cool technique.

Relaxation oscillations

Let's write the reduced Oregonator again and define a couple of extra functions

$$\epsilon \dot{u} = g_1(u, w) = u - u^2 - fw\frac{u - \mu}{u + \mu}, \tag{71a}$$

$$\dot{w} = g_2(u, w) = u - w. \tag{71b}$$

Recall that $\epsilon \approx 10^{-2}$, which means that u is moving at a considerably faster pace than w. You can sort of see that in the time series of Figure 14. Here w moves at a consistent pace but u (comparatively) zooms to and from 0 to 0.9 in a very small interval of time. Notice

that the oscillation seems to be structured around the inflexion points of the nullcline N_u. This is not a coincidence.

We mentioned earlier that the value of ϵ implies that u evolves at a faster pace than w, *if the right-hand sides of the equations are of a similar magnitude*. When the oscillation approaches the nullcline, then $\dot{u} \approx 0$ and u slows down to the point that it and w evolve at a similar pace. We can hold on to this fact to break down the oscillations into four phases, which follow the labelling of Figure 15, starting at the orange point (A)

- A: The trajectory follows the nullcline in counterclockwise direction. It cannot move too far away from N_u because, if it does, then $\dot{u} \gg \dot{w}$ and the orbit moves horizontally towards N_u again.
- B: This behaviour continues until the trajectory reaches the maximum of the nullcline. $\dot{u}$ is still negative, which keeps driving it to the left. However, without the nullcline to slow it down, the trajectory zooms in an almost horizontal movement until it reaches C.
- C: Here the trajectory once again reaches the nullcline (except on the other side) and it follows this curve downwards (according to the phase plane) until reaching D.
- D: The flow now pushes the orbit to the right, and without the nullcline to stop the trajectory, u increases in a very short amount of time, returning to A.

What we just described is known as a *relaxation oscillation*, and it depends on the difference in timescales between the fast variable (u) and the slow one (w). As $\epsilon \to 0$ the computed trajectory will keep approaching the ABCDA cycle. This description is a very crude attempt to perform a technique called **geometric singular perturbation theory** (GSPT). We won't get into the details of GSPT in this course. One of the advantages of invoking such a theory is that it guarantees that, for a range of small non-zero values of ϵ the behaviour we just described is still valid, without having to prove it ourselves.

The other enormous advantage of employing theory in this manner is that it allows us to describe the behaviour of the oscillation, even when we lack a closed-form solution for it. Remember, this is all *qualitative* analysis.

The efficacy of the Oregonator

To finish out little journey into chemical oscillations, let us bring everything that we have learned so far together and use the Oregonator to describe the oscillations of the BZ reaction. To do that we must first undo our nondimensionalisation and return to the original variables (after all, the experimental data in those units).

Figure 16 shows the trajectories generated by the simplified Oregonator model (with the appropriate choice of parameters) alongside a particularly nice image obtained by the authors of the FKN mechanism as they attempted to replicate the BZ reaction in their lab. In both images, the first time series shows the Ce^{4+} concentration, and the second one Br^-.

Go back to the original FKN mechanism, and behold the level of complexity involved in those 10 equations. The beauty of the simplified Oregonator lies in its capacity to reproduce such complex behaviour with only two variables and a set of (relatively) simple ODEs. Sure, it is not a 'one-to-one' replica (the cerium oscillations have a different bent in the experimental data), but *qualitatively* it is a perfect match.

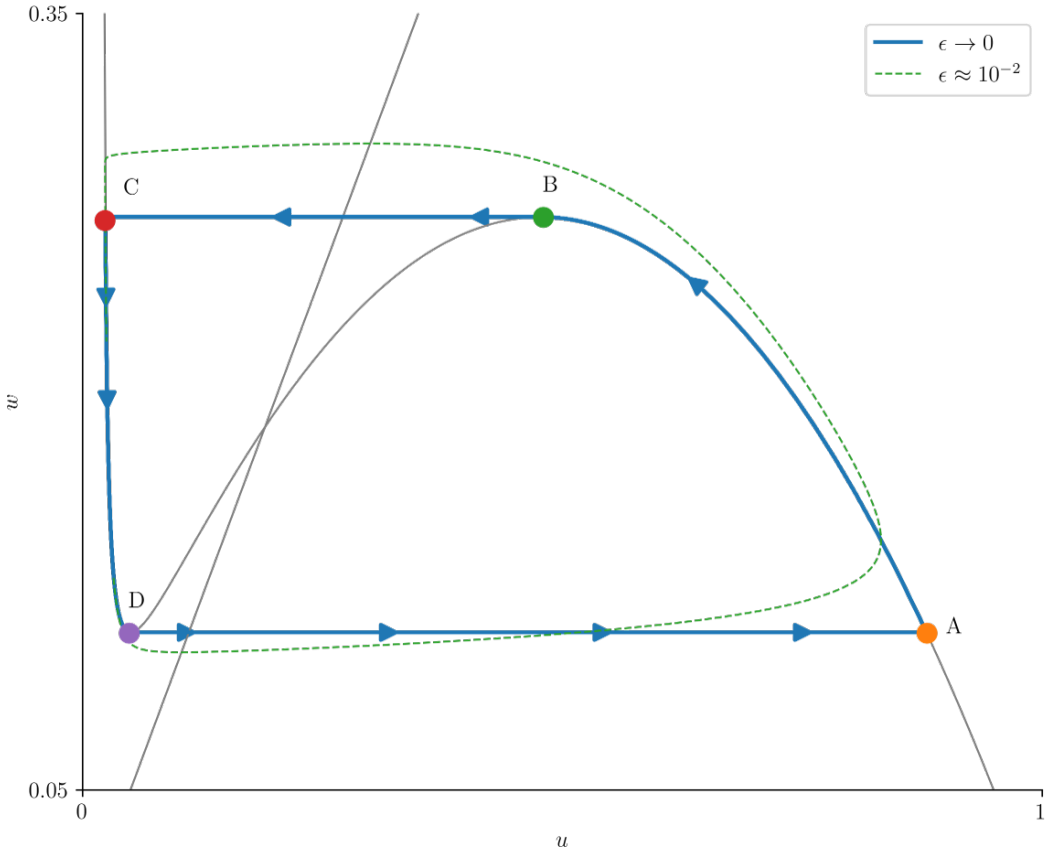


Figure 15: A close up of the relaxation oscillation exhibited by the simplified Oregonator. In this example we have decreased the value of ϵ to make the difference in timescales between u and w more noticeable.

And the other great advantage of using such a simplified model is we can now finally describe in a few words the core of the mechanism that upended the field of chemical kinetics in the XX century. Our final diagram, Figure 17, shows the phase plane with the relaxation oscillation in the original variables. To finish, we will construct a description of the oscillating mechanism as follows:

- Starting at $[Br^-]_{min}$, the concentration of Br^- (the autocatalyst) rises slowly alongside that of Ce^{4+}, until it reaches $[Br^-]_{jump}$.
- At $[Br^-]_{jump}$, the concentration of bromous acid $[HBrO_2]$ decreases sharply as it is all being transformed into Br^-.
- Once the majority of $HBrO_2$ has been consumed ($[Br^-]_{max}$), the oscillation changes phase and the concentration of both Ce^{4+} and Br^- starts to decay slowly.
- As the levels of Br^- reach a critical value ($[Br^-]_{crit}$), this compounds gets rapidly transformed back into bromous acid. This behaviour continues until the majority of the Br^- is consumed, and the concentration reaches is minimal value ($[Br^-]_{min}$).

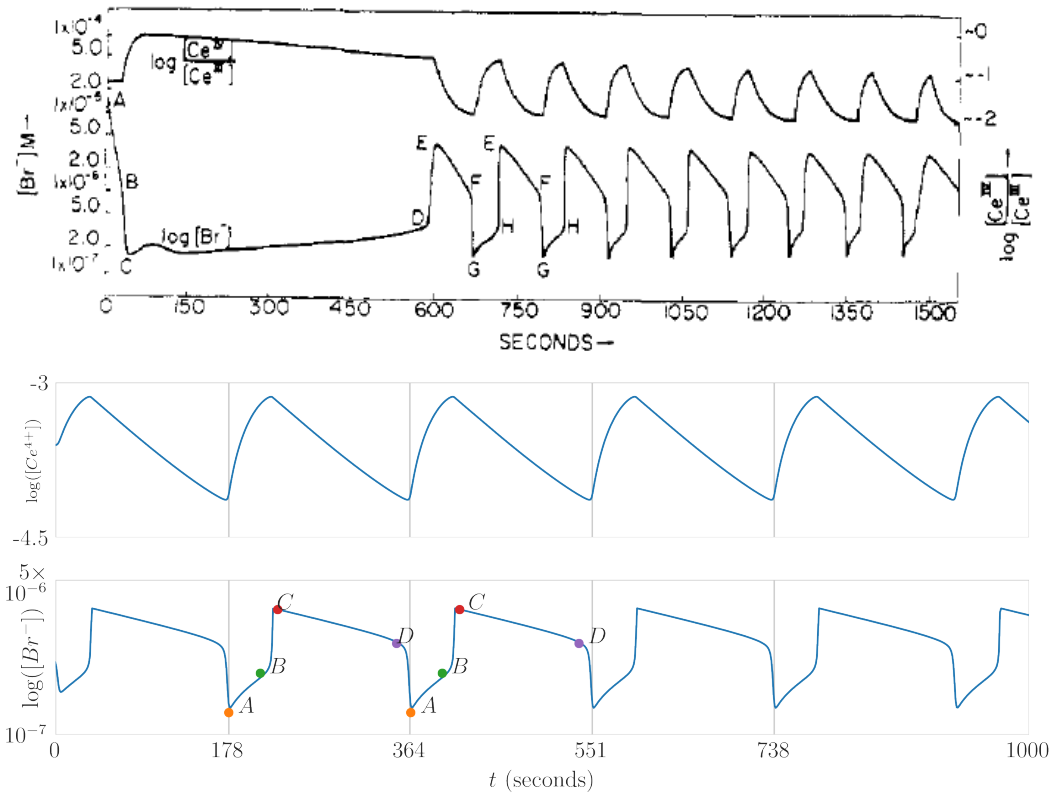


Figure 16: The top panel shows the oscillations of the BZ reaction as observed by Field, Kőrös and Noyes Field et al (1972). The bottom panel shows trajectories generated by the simplified Oregonator model (in their original dimensional form), with the A, B, C and D points from the relaxation analysis highlighted in the curve.

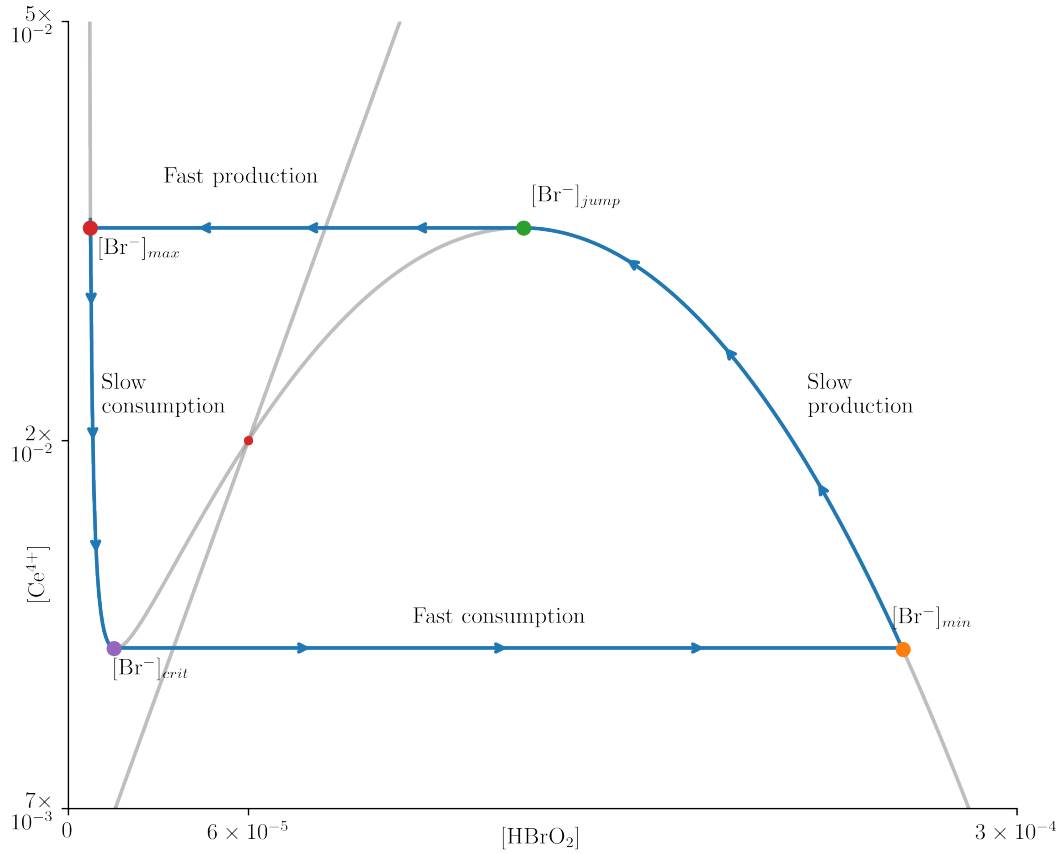


Figure 17: A close up of the relaxation oscillation exhibited by the simplified Oregonator, using the same parameter values as the previous figure.

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